

BANK SECURE NETWORK

Secure Network Design and Simulation using Cisco Packet Tracer

Technical Documentation

Author: Andrew D Souza

Project Type: Networking & Cybersecurity

Tools: Cisco Packet Tracer

GitHub Repository: bank-secure-network

1. Project Overview

Bank Secure Network is a simulated banking network designed and implemented using Cisco Packet Tracer. The project demonstrates VLAN segmentation, inter-VLAN routing, centralized DHCP and DNS services, an internal banking web service, and ACL-based access control.

The network separates different types of users and systems into dedicated VLANs, including administrators, employees, ATMs, and servers. A Cisco router provides communication between these network segments, while DHCP and DNS services provide centralized network configuration and name resolution.

Network security is implemented using Access Control Lists (ACLs) to restrict unauthorized communication between selected network segments while allowing legitimate access to critical banking services.

Key Technologies

Technology	Purpose
Cisco Packet Tracer	Network design, configuration, and simulation
VLAN	Segments the network into separate logical networks
Router-on-a-Stick	Enables communication between different VLANs
DHCP	Automatically assigns IP configuration to clients
DNS	Resolves <code>banking.local</code> to the Banking Server
ACL (Access Control List)	Restricts unauthorized communication between VLANs
HTTP/HTTPS	Provides access to the internal banking web service
IPv4	Provides addressing and communication across the network

2. Problem Statement

Banking networks handle different types of users, devices, and critical services. If these systems operate on a flat network, unauthorized access between departments and devices can become easier, increasing the risk of lateral movement and exposure of sensitive resources.

This project addresses this problem by designing a segmented banking network in Cisco Packet Tracer. Administrators, employees, ATMs, and servers are placed in separate VLANs, with controlled communication between them. Network services such as DHCP and DNS are centralized, while ACLs are used to restrict unauthorized traffic between network segments.

Project Goals

- Separate users and systems using VLANs.
- Enable controlled communication between different network segments.
- Provide centralized DHCP and DNS services.
- Host an internal banking web service.
- Restrict unauthorized inter-VLAN communication using ACLs.
- Test and verify the implemented security policies.

3. Network Architecture

The banking network is designed using a hierarchical structure consisting of a central router, a core switch, dedicated access switches, end devices, and servers. The router provides inter-VLAN routing, while the switches provide connectivity and VLAN-based segmentation.

The network is divided into four logical segments: ADMIN, EMPLOYEE, ATM, and SERVER. This separation helps isolate different categories of devices and allows traffic between segments to be controlled through the router.

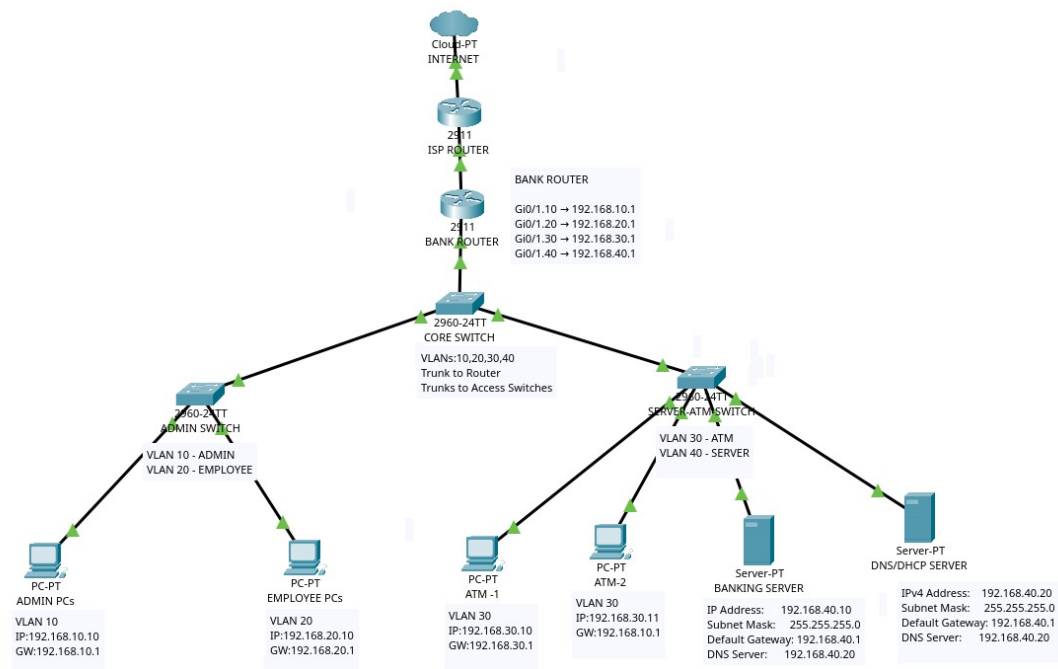


Figure 1: Bank Secure Network Topology

Network Components

Component	Quantity	Purpose
Cisco 2911 Router	1	Inter-VLAN routing and traffic control
Cisco 2960 Switch	3	Device connectivity and VLAN segmentation
Banking Server	1	Hosts the internal banking web service
DNS/DHCP Server	1	Provides centralized DNS and DHCP services
Admin PC	1	Represents administrative users
Employee PC	1	Represents bank employees
ATM	2	Represents ATM devices
Internet Cloud	1	Represents external network connectivity

4. VLAN & IP Addressing

VLANs were used to divide the banking network into separate logical segments based on device and user roles. This reduces unnecessary broadcast traffic and provides a foundation for controlling communication between different parts of the network.

VLAN	Name	Network	Default Gateway	Devices
10	ADMIN	192.168.10.0/24	192.168.10.1	ADMIN-PC
20	EMPLOYEE	192.168.20.0/24	192.168.20.1	EMPLOYEE-PC
30	ATM	192.168.30.0/24	192.168.30.1	ATM-1, ATM-2
40	SERVER	192.168.40.0/24	192.168.40.1	BANKING-SERVER, DNS/DHCP-SERVER

Server Addressing

Server	IP Address	Gateway	DNS
BANKING-SERVER	192.168.40.10	192.168.40.1	192.168.40.20
DNS/DHCP-SERVER	192.168.40.20	192.168.40.1	192.168.40.20

5. VLAN Configuration

VLANs were created on the switches to logically separate administrative users, employees, ATMs, and servers. Each VLAN was assigned a unique VLAN ID and name according to the network design.

End devices were connected to access ports configured for their respective VLANs. Trunk links were configured between switches and between the core switch and router. Trunking allows traffic from multiple VLANs to travel across a single physical link using IEEE 802.1Q VLAN tagging.

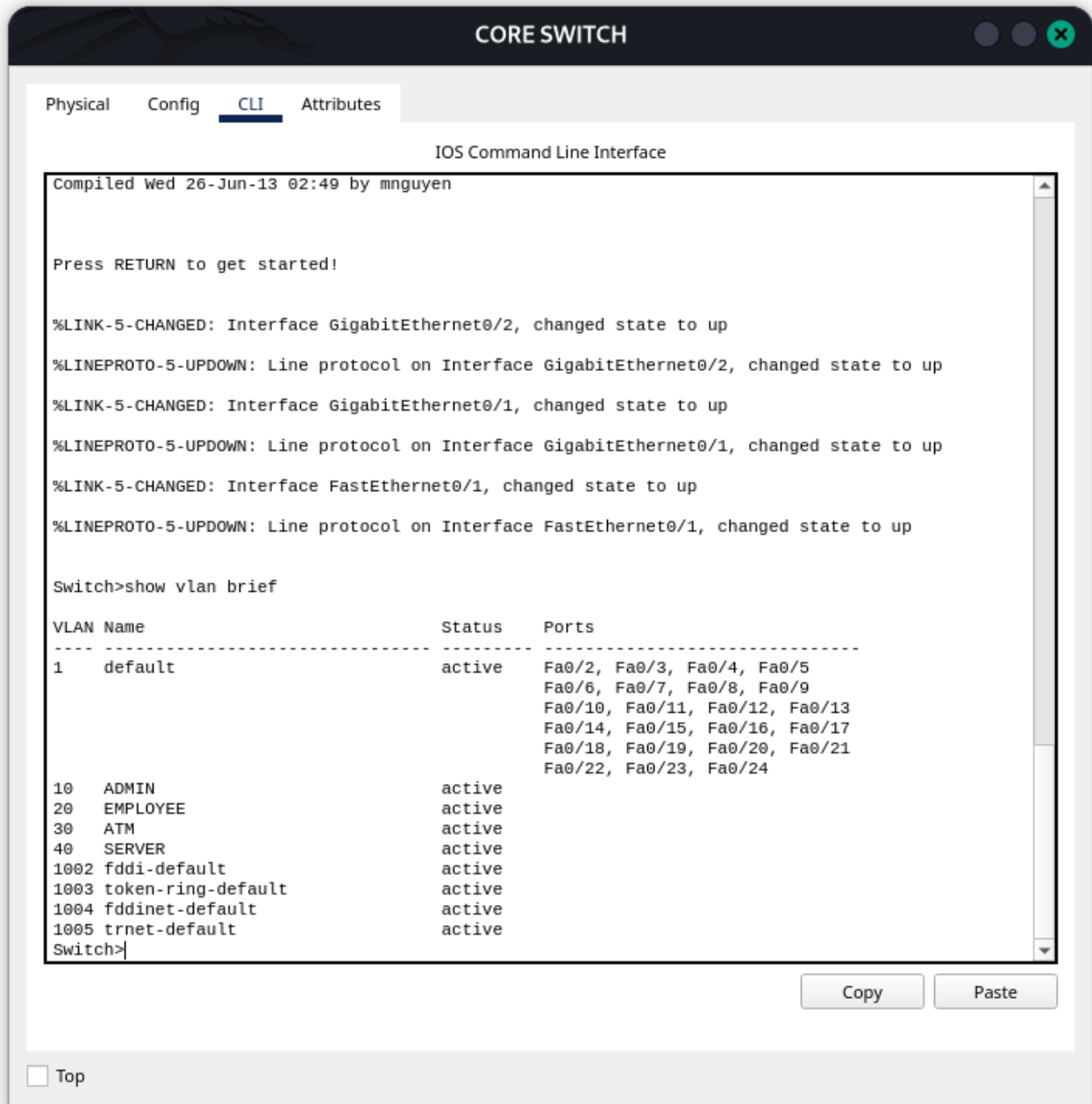


Figure 2: VLAN configuration and access-port assignments

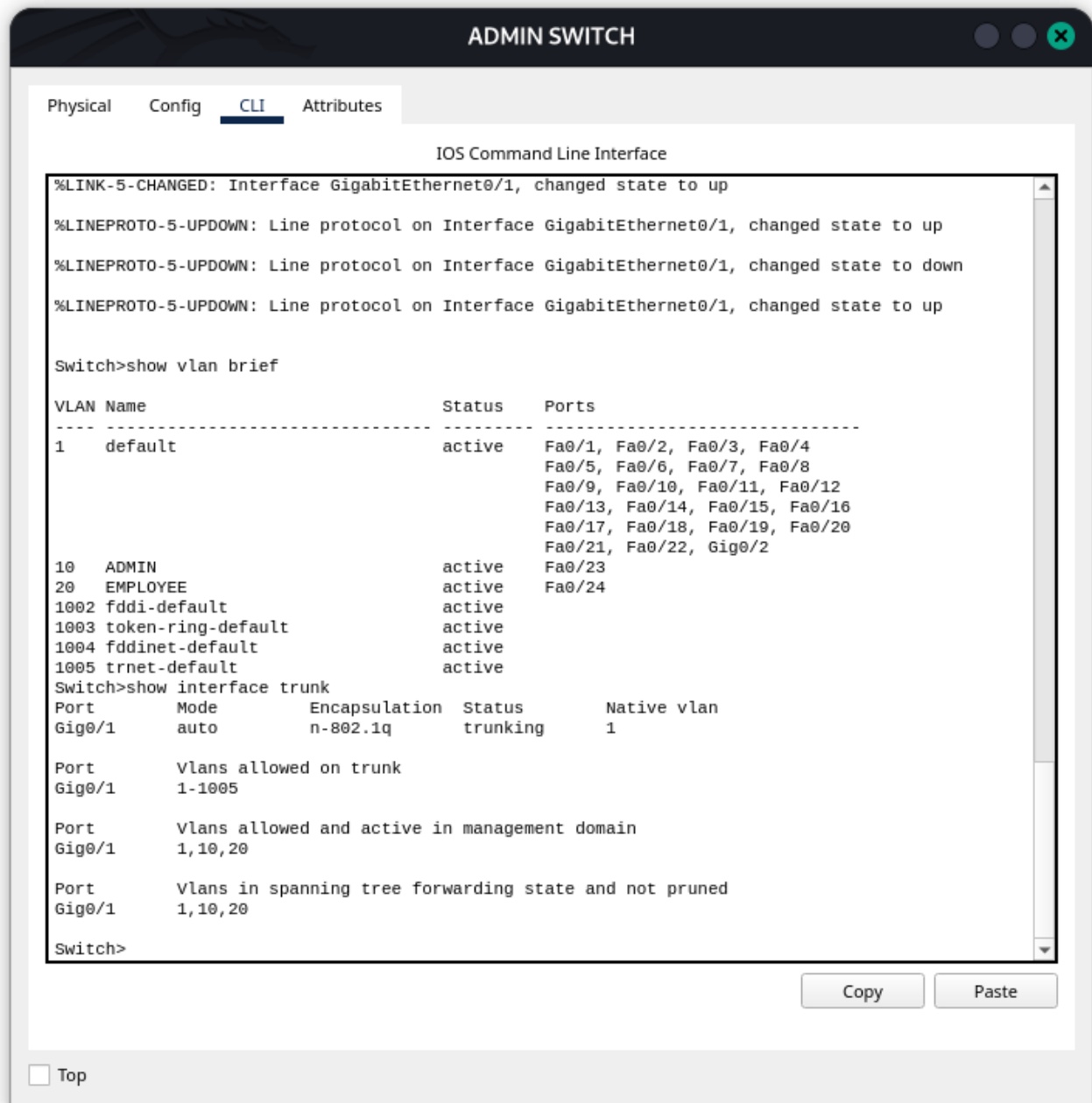


Figure 3: Trunk links carrying multiple VLANs

6. Inter-VLAN Routing

Router-on-a-stick is used to provide communication between the four VLANs. The router's GigabitEthernet0/1 interface is divided into 802.1Q subinterfaces, with one subinterface acting as the default gateway for each VLAN.

Subinterface	VLAN	Gateway
Gi0/1.10	10	192.168.10.1
Gi0/1.20	20	192.168.20.1
Gi0/1.30	30	192.168.30.1
Gi0/1.40	40	192.168.40.1

7. DHCP Configuration

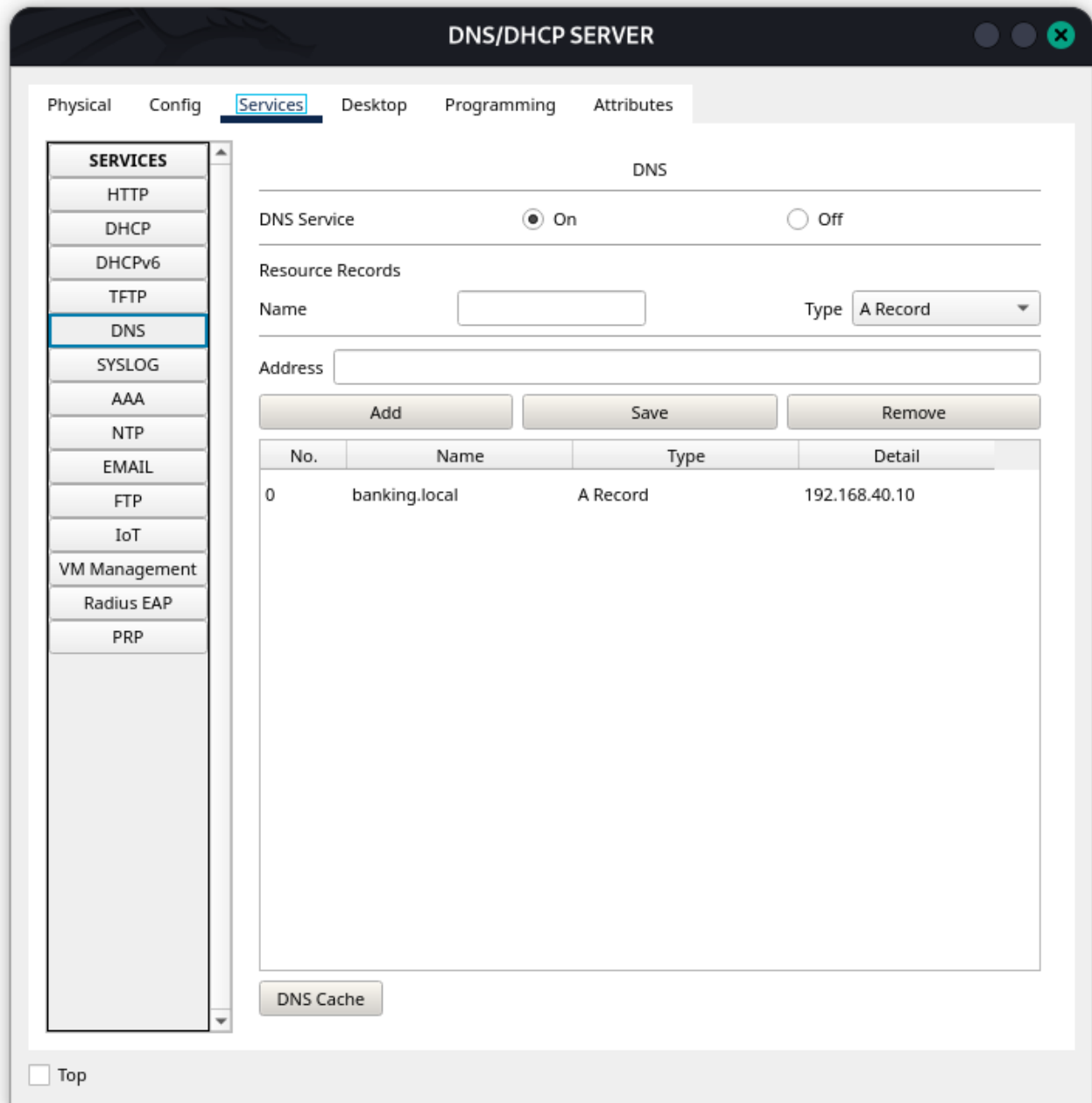
A centralized DHCP server at 192.168.40.20 provides automatic IP configuration to client devices. Separate DHCP pools are used for the ADMIN, EMPLOYEE, and ATM networks.

Because the DHCP server is located in VLAN 40, DHCP relay is configured on the router using ip helper-address 192.168.40.20 for the client VLANs.

The screenshot displays the configuration page for the ATM-1 interface on a network device. The 'Desktop' tab is selected under the 'Config' section. The 'IP Configuration' window is open for the 'FastEthernet0' interface. The 'DHCP' option is selected for IP configuration, with the following settings: IPv4 Address (192.168.30.10), Subnet Mask (255.255.255.0), Default Gateway (192.168.30.1), and DNS Server (192.168.40.20). The 'IPv6 Configuration' section shows the 'Static' option selected, with fields for IPv6 Address, Link Local Address (FE80::260:5CFF:FE69:8ACA), Default Gateway, and DNS Server. The '802.1X' section is also visible, with 'Use 802.1X Security' unchecked, 'Authentication' set to 'MD5', and fields for Username and Password. A 'Top' link is located at the bottom left of the configuration area.

8. DNS Configuration

The DNS service runs on 192.168.40.20. A DNS record maps the hostname banking.local to the Banking Server at 192.168.40.10, allowing clients to access the service using a hostname instead of an IP address.



The screenshot shows the 'DNS/DHCP SERVER' configuration window. The 'Services' tab is selected in the top navigation bar. On the left, a list of services includes HTTP, DHCP, DHCPv6, TFTP, DNS (highlighted), SYSLOG, AAA, NTP, EMAIL, FTP, IoT, VM Management, Radius EAP, and PRP. The main area is titled 'DNS' and contains the following configuration options:

- DNS Service:** A radio button interface with 'On' selected and 'Off' unselected.
- Resource Records:** A section for adding and managing DNS records.
- Name:** A text input field.
- Type:** A dropdown menu currently set to 'A Record'.
- Address:** A text input field.
- Buttons:** 'Add', 'Save', and 'Remove' buttons are located below the input fields.
- Table:** A table displaying the current DNS records.
- DNS Cache:** A button located at the bottom of the configuration area.

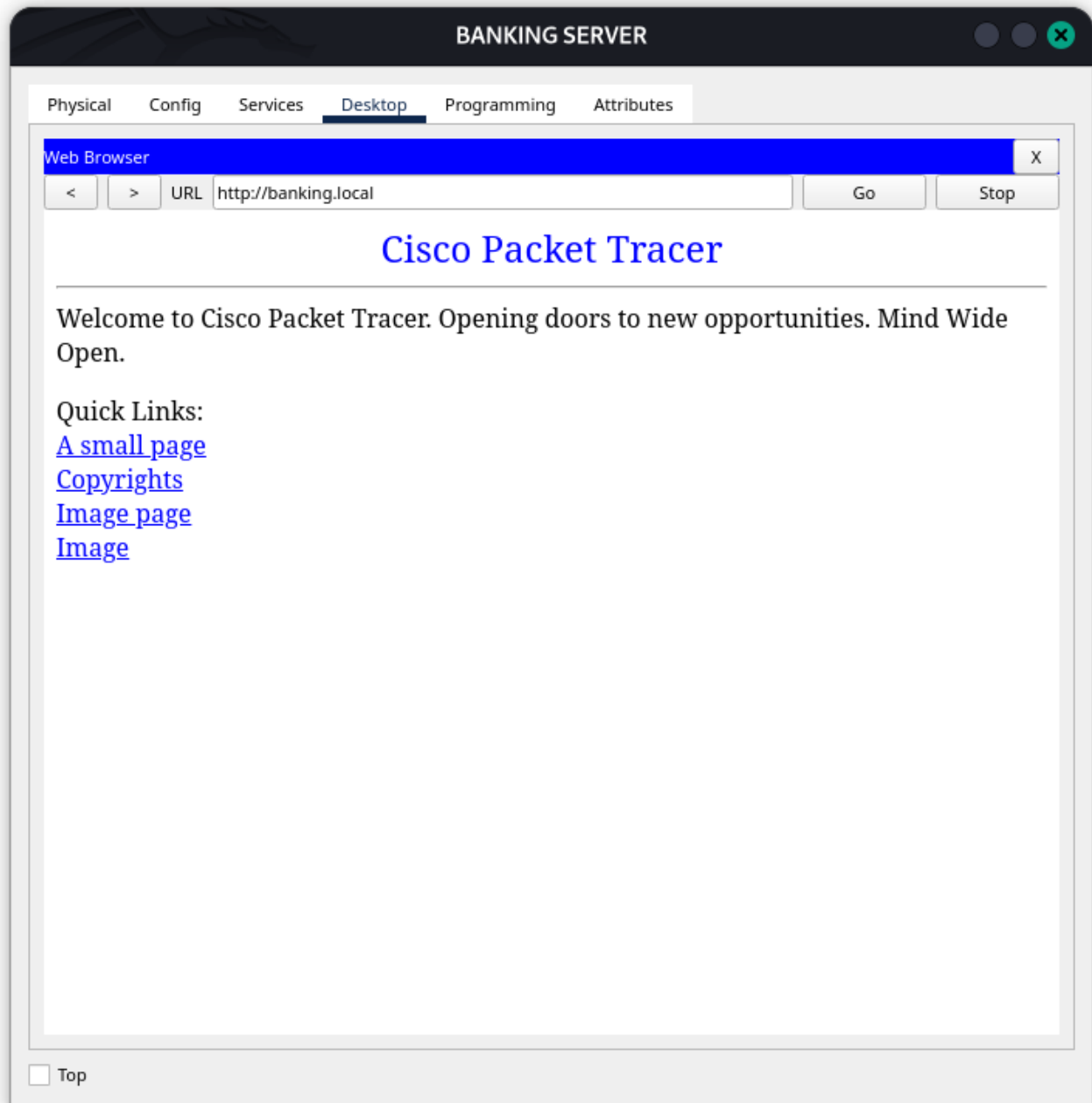
No.	Name	Type	Detail
0	banking.local	A Record	192.168.40.10

At the bottom left of the window, there is a 'Top' button.

9. Banking Web Server

The Banking Server uses the static address 192.168.40.10/24 with 192.168.40.1 as its default gateway. HTTP and HTTPS services are enabled to provide an internal banking web service.

The service can be accessed using `http://banking.local` after DNS resolution.



BANKING SERVER

Physical

Config

Services

Desktop

Programming

Attributes

IP Configuration

X

IP Configuration

☐ DHCP

☒ Static

IPv4 Address

192.168.40.10

Subnet Mask

255.255.255.0

Default Gateway

192.168.40.1

DNS Server

192.168.40.20

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address

/

Link Local Address

FE80::204:9AFF:FE99:E843

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication

MD5

Username

Password

☐ Top

10. ACL-Based Network Security

Access Control Lists were implemented on the router to restrict communication between selected VLANs. The rules are applied inbound on the relevant router subinterfaces.

ACL	Restriction	Allowed Traffic
EMPLOYEE-SECURITY	Employee → Admin	Other required traffic, including Banking Server access
ATM-SECURITY	ATM → Admin; ATM → Employee	Other required traffic, including Banking Server access

The ACL design follows a simple segmentation principle: users and ATM devices cannot directly communicate with restricted internal networks, while required access to the banking server and network services remains available.

11. Testing & Verification

The network was tested using ping, DNS resolution, web access, DHCP verification, and ACL match counters. Both permitted and blocked communication paths were tested to confirm that the design and security policies were functioning as intended.

Test	Source → Destination	Expected Result	Result
DHCP	EMPLOYEE-PC → DHCP	Receives 192.168.20.x configuration	PASS
DHCP	ATM-1 → DHCP	Receives 192.168.30.x configuration	PASS
DNS	EMPLOYEE-PC → banking.local	Resolves to 192.168.40.10	PASS
Web	EMPLOYEE-PC → banking.local	Banking page loads	PASS
ACL	EMPLOYEE-PC → Admin	Traffic blocked	PASS
ACL	ATM-1 → Admin	Traffic blocked	PASS
ACL	ATM-1 → Employee	Traffic blocked	PASS
Connectivity	ATM-1 → Banking Server	Traffic allowed	PASS

12. Security Features

- VLAN-based network segmentation.
- Controlled inter-VLAN routing through router-on-a-stick.
- ACL-based restrictions between selected network segments.
- Centralized DNS and DHCP services.
- Dedicated server VLAN for critical services.
- Verification through both allowed and blocked traffic tests.

13. Limitations

This project is a Cisco Packet Tracer simulation and does not represent all controls found in a production banking environment. A real banking network would require additional controls such as enterprise firewalls, IDS/IPS, endpoint security, strong authentication, monitoring, redundancy, and formal security policies.

14. Future Improvements

- Deploy dedicated firewalls and more advanced traffic inspection.
- Add IDS/IPS and centralized security monitoring.
- Introduce stronger authentication and role-based access.
- Add redundancy for critical network devices and services.
- Implement VPN-based secure remote access where required.

15. What I Learned

This project provided practical experience with VLAN segmentation, trunking, router-on-a-stick, DHCP relay, DNS, ACL configuration, IP addressing, connectivity testing, and troubleshooting. It also demonstrated how network design decisions affect both connectivity and security.

16. Conclusion

The Bank Secure Network successfully demonstrates a segmented and controlled banking network in Cisco Packet Tracer. VLANs separate different device and user groups, router-on-a-stick enables controlled inter-VLAN communication, centralized DHCP and DNS provide essential network services, and ACLs restrict unauthorized traffic. Testing confirmed that required services remain accessible while selected communication paths are blocked.

17. Appendix – Key Configurations

Add selected router and switch configurations here, such as VLAN creation, trunk configuration, router subinterfaces, DHCP relay, and ACL rules. Include only the commands that are useful for understanding or reproducing the project.